Title: Embodied Teleoperation and Imitation Learning for Human-like Robotic Manipulation

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Aim: To develop a real-time telemanipulation testbed in which a human operator controls a robotic arm via a sensorised glove or AR/VR interface with depth-based tracking. The system will prioritise low-latency, stable embodied control, while systematically evaluating how increasing communication delay (<5secs) affects performance in contact-rich manipulation tasks. The platform will additionally generate structured demonstration data for imitation learning, enabling future research in autonomous and shared robotic control.

Relevance: Teleoperation enables skilled human control in environments that are hazardous, remote, or costly to access, including orbital operations, subsea intervention, and nuclear decommissioning. However, performance degrades under communication delay, limiting effectiveness in real-world deployment. This project addresses both operational and research challenges. First, it quantifies how delay impacts manipulation performance in a controlled setting. Second, it reframes teleoperation as a data generation mechanism for learning-based robotics, where each session produces synchronised state-action trajectories suitable for imitation learning and shared autonomy. This aligns with emerging trends in embodied AI and human-in-the-loop robotic learning.

Novelty: Teleoperation systems and latency compensation are both well studied individually, but there is no widely available dataset characterising how a glove-plus-camera teleoperation setup, with and without haptic feedback, performs across a controlled latency sweep on a standardised contact-rich benchmark. The contribution is this dataset together with a reproducible benchmark protocol, which other groups can use to evaluate delay-compensation algorithms, shared autonomy controllers, and learned teleoperation policies.

Method and Work Plan: A robotic manipulator is controlled via depth-based skeletal tracking and a sensorised glove (or AR/VR interface), mapping human motion to robot end-effector pose and gripper actuation. A ROS2-based pipeline integrates a configurable latency injection layer. Tasks are based on a subset of the YCB object set, enabling reproducibility and comparison. The project is structured as follows: **[Weeks 1–2]: System Setup and Hardware Integration.** Integration of depth camera, skeletal tracking pipeline, and initial robot interface. Setup of glove or AR/VR input for hand pose and finger tracking. **[Weeks 3–4]: Control and Mapping Development.** Implementation of human-to-robot motion mapping (end-effector control, inverse kinematics, gripper actuation). Development of a stable control loop for teleoperation. **[Weeks 5–6]: ROS2 Integration for Real Time Implementation.** Development of ROS2 communication pipeline and configurable latency module (<5secs). Initial system validation under nominal conditions. **[Weeks 7–8]: Testing and Experimental Benchmarking.** Execution of contact-rich pick-and-place tasks across latency conditions and haptic feedback modes. Data logging of task performance (completion time, success rate, contact force, NASA-TLX workload). **[Weeks 9–10]: Data Analysis and Learning Extension.** Processing of collected dataset into state-action pairs for imitation learning. Preliminary evaluation of demonstration quality under delay and short analysis of performance degradation trends.

Feasibility: The project is feasible within 10 weeks due to its modular design and use of existing components. It builds on ongoing teleoperation work at NR, reducing integration risk. Off-the-shelf tools, including ROS2 drivers, depth camera tracking SDKs, and the YCB object set, streamline development, with real-time control prioritised and delay treated as an experimental variable. **NR Equipment Required**: Access to RAS lab, one robotic arm (xArm) for the duration of the project, one depth camera, and workstation access for ROS2 development. The glove or headset, robotic hand end-effector, and haptic feedback hardware will be built by the student.